Hence

The block's friction force on the belt has magnitude

It points along . In the ground frame, belt speed is . The block's work on the belt per unit time is

Thus the additional power required by the load is

The motor's no-load output power is

Its output power with cargo is

Let be the operating current with cargo. Then

Solving gives

Scoring (points): ; total ; ; ; ; ; ; .

III. (points)

The light rod has length . At angle to the horizontal, balls , have speeds , . The rod's angular speed is . Conservation of mechanical energy gives

Also,

Solving gives